FD-55F Series

Technical Specifications

			FD-55A		FD-55B		FD-55E		FD-55F**		FD-55G***	
			FM	MFM	FM	MFM	FM	MFM	FM	MFM	FM	MFM
Transfer Rate (K bits/sec)			125	250	125	250	125	250	125	250	250	500
Capacity (K bytes)	Unformatted	Per Track	3.125	6.25	3.125	6.25	3.125	6.25	3.125	6.25	5.028	10.416
		Per Disk	125	250	250	500	250	500	500	1,000	802	1,604
	Formatted (16 sectors/ track)*	Per Sector	0.128	0.256	0.128	0.256	0.128	0.256	0.128	0.256	0.256	0.512
		Per Track	2.048	4.096	2.048	4.096	2.048	4.096	2.048	4.098	3.840	7.680
		Per Disk	81.92	163.84	163.84	327.68	163.84	327.68	327.68	655.36	591	1,183
Inside Track Recording Density (bpi)			2768	5536	2938	5876	2788	5578	2961	5922	4823	9646
Inside Track Flux Density (frpi)			5536		5876		5576		5922		9646	
Tracks/Disk			40		80		80		160		154	
Track Density (tpi)			48				96				2x 77 tracks	
Track Radius (mm)		Outside	57.150		side 0 : 57.150 1 : 55.033		57.150		side 0 : 57.150 1 : 55.033			
		Inside	36.513		side 0 : 36.513 1 : 34.396		36.248		side 0 : 36.248 1 : 34.131		side 0 : 37.042 1 : 34.925	
Average Access Time (ms)			93				94				91	
Track Access Time (ms)			6				3					
Setting Time (ms)			15									
Disk Rotational Speed (rpm)			300								360	

*FD-55G: 15 sectors/track

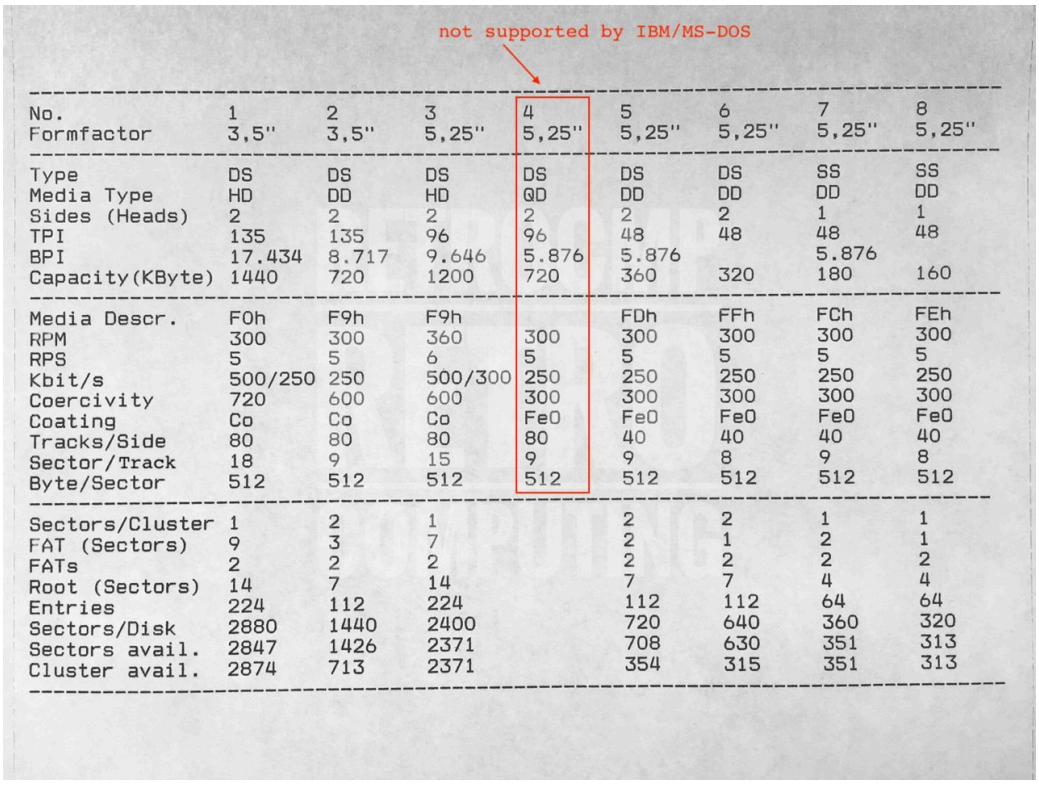
**Also FD-55GF Normal Density ***Also FD-55GF High Density
Features and specifications are subject to change without notice.

TEAC FD55 technical specifications *[Brochure TEAC FD-55 Series]*

The TEAC FD-55F never appears in the world of IBM compatible computers. Why is that? Because it is simply superfluous in the DOS world.

The TEAC FD-55F is a double-sided 5.25 inch drive that can write 96 tracks per inch (TPI) with 9 sectors per track. In other words, the capacity of a floppy disk is 720 KByte. This corresponds to the capacity of the standard 3.5-inch double-density (DD) drive. This 3.5-inch format has been supported since DOS version 3.2.

This was probably one reason why the 5.25 drives with 720 KByte did not find their way into the IBM world.



not supported by IBM/MS-DOS

No.	1	2	3	4	5	6	7	8
Formfactor	3,5"	3,5"	5,25"	5,25"	5,25"	5,25"	5,25"	5,25"
Type	DS	DS	DS	DS	DS	DS	SS	SS
Media Type	HD	DD	HD	QD	DD	DD	DD	DD
Sides (Heads)	2	2	2	2	2	2	1	1
TPI	135	135	96	96	48	48	48	48
BPI	17.434	8.717	9.646	5.876	5.876		5.876	
Capacity(KByte)	1440	720	1200	720	360	320	180	160
Media Descr.	F0h	F9h	F9h		FDh	FFh	FCh	FEh
RPM	300	300	360	300	300	300	300	300
RPS	5	5	6	5	5	5	5	5
Kbit/s	500/250	250	500/300	250	250	250	250	250
Coercivity	720	600	600	300	300	300	300	300
Coating	Co	Co	Co	FeO	FeO	FeO	FeO	FeO
Tracks/Side	80	80	80	80	40	40	40	40
Sector/Track	18	9	15	9	9	8	9	8
Byte/Sector	512	512	512	512	512	512	512	512
Sectors/Cluster	1	2	1		2	2	1	1
FAT (Sectors)	9	3	7		2	1	2	1
FATs	2	2	2		2	2	2	2
Root (Sectors)	14	7	14		7	7	4	4
Entries	224	112	224		112	112	64	64
Sectors/Disk	2880	1440	2400		720	640	360	320
Sectors avail.	2847	1426	2371		708	630	351	313
Cluster avail.	2874	713	2371		354	315	351	313

Floppy disk formats supported by IBM compatible computers

> *The MS-DOS Enzyclopedia, 1988, p. 865*
>
> *FORMAT [drive] /4 - formats a standard double sided, double density disk (360 KB) on a quad-density disk drive.*
>
> *Note: Each OEM determines which switches will be supported by the FORMAT utility included with the versions of MS-DOS sold with its computers.*

I don't know any floppy disk controller or standard BIOS that supports this 5.25 inch QD disk drives on IBM compatible computers. Also the CompatiCard only supports a pseudo format with 800 KByte but not 720 KByte.

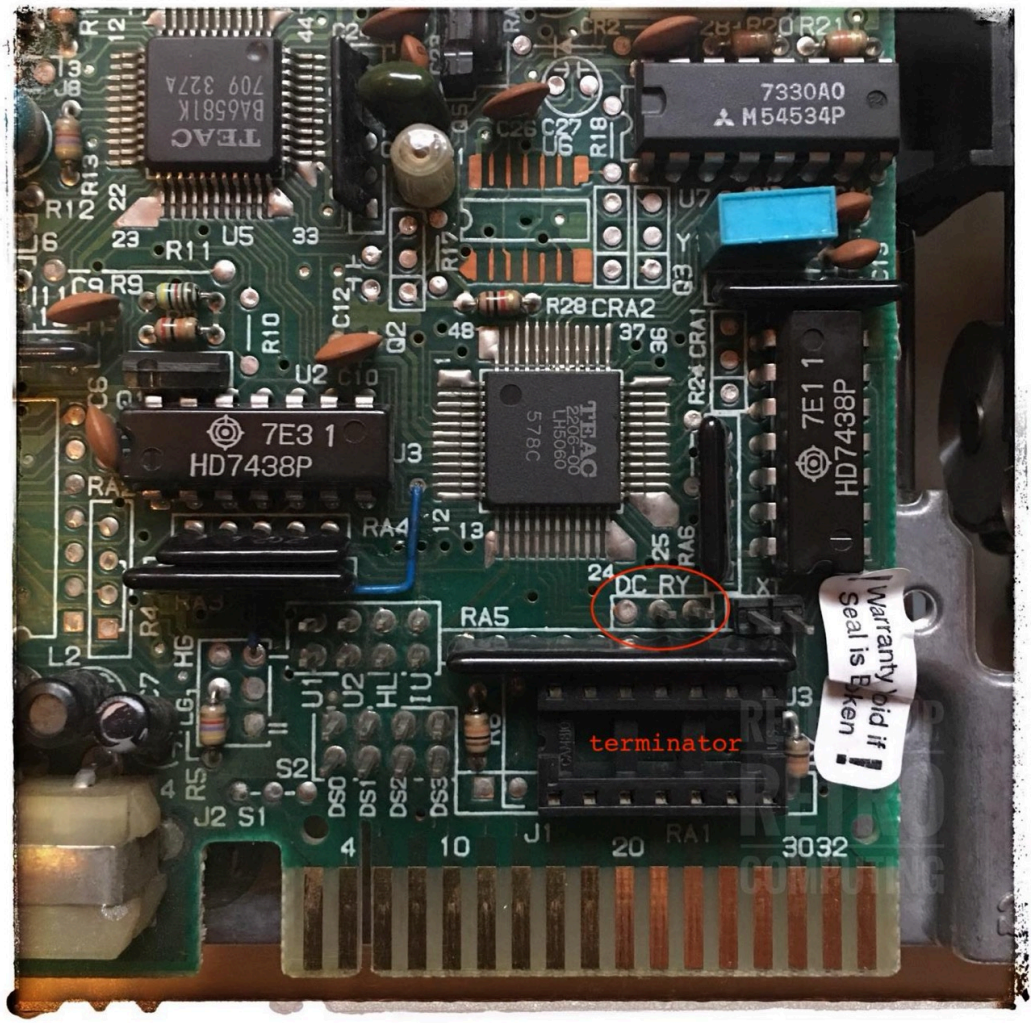
In the CP/M world the TEAC FD-55F can be found very often. There are a lot of disk formats with 80 tracks or 96 TPI. You can easily test this with the DOS program 22DISK. Simply select the appropriate format and format the diskette (initialize). This works without any problems. You even get a hint from 22DISK that 96 TPI (5.25 inch) are only supported by 1.2 MByte drives or by the TEAC FD-55F!

Jumper Settings

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You have to be a little careful with the jumper settings. I have for example the type FD-55F-30. This has no DISK CHANGE (DC), but only

the READY (RY) jumper. In other words, this drive does not work on an AT (I also tested it), but on a PC/XT, see also under Jumper settings. Another special feature is the XT jumper, which I don't use, I only use RY.



TEAC FD-55F Jumper (RY)

My jumper settings are: DS1 / IU / RY / FG

On the picture above you can also see the so called terminator or resistor pack. It is not set because I am using the TEAC FD-55F as drive B:. Only the last drive on the cable must have a terminating resistor installed! Attention! The TEAC FD-55F requires a 14 PIN resistor with 330 Ohm!

A Test With 22Disk and Uniform

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I use the TEAC FD-55F with my IBM PC or XT and it works fine in combination with the programs 22Disk und Unifom PC. In other words, I use the floppy drive only in conjunction with external CP/M formats. Example:

22Disk:

1. c:> cfmt /kay4 b:

2. c:> dtoc /kay4 test.txt b:

The first command formats a diskette in drive B: in CP/M format DS, DD, 96 tpi (kay4) for the Kaypro Pro-8 Rom. The second command copies the file test.txt to drive B:.

With Uniform I selected the CP/M format "Kaypro Pro-8 Rom DSDD:96:CPM". Under PC-DOS the access is quite simply possible with DIR B:. The file test.txt is displayed correctly.

Changing a FD-55GFR to a FD-55F

top

... or, how do I get my drive to spin at 300 RPM instead of 360?

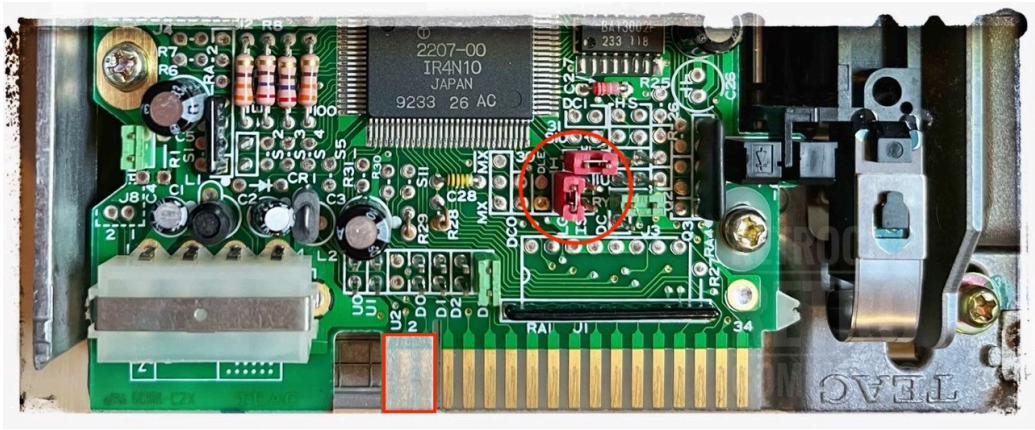
You need a 720 KByte floppy drive for your CP/M Kaypro but you only have a FD-55GFR (1.2 MByte)! Nothing easier than that. Just reduce the rotation speed from 360 RPM to 300 RPM and you have a 720 KByte drive with 80 tracks and 96 TPI (double sided, quad density, DSQD).

This modification can be done quite easily with the following four TEAC drives.

- TEAC FD-55GFR-149-U
- TEAC FD-55GFR-193-U
- TEAC FD-55GFR-570-U
- TEAC FD-55GFR-7193-U

I have tested this with my versions -149 and -193. It works flawlessly. There are many other subtypes of the FD-55GFR. I therefore assume that the situation will be similar for the others. Just give it a try.

Decisive here are the three jumpers I, IS and LG, see TEAC FD-55 jumper settings below. Furthermore, you need to tape contacts 1, 2, 3 and 4, that's it. You can also find out more details here.



TEAC FD-55GFR - 300 RPM mod

Hearing test: Use a jumper with a long grip for your test at "I". Make the drive rotate and remove the jumper. If you have good hearing, then you can "hear" how the rotation speed changes.

Specifications (S) and Brochures (B)

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- S: TEAC FD-55FV
- S: TEAC FD-55 Jumper Settings
- B: TEAC FD-55 Series (A, B, E, F, G, GF)

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